

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION TO THE PLANNING PHASE The Planning Phase establishes the foundational blueprint for developing the ComicCraft AI platform. Given the technical intricacies of linking a local FastAPI server with multiple external cloud-based Generative AI models (Google Gemini and Hugging Face Stable Diffusion), a phased, strategic roadmap was essential to guarantee a smooth development, testing, and deployment cycle.

3. ITERATIVE DEVELOPMENT STRATEGY Rather than a traditional waterfall approach, the project utilized an iterative, sprint-based workflow. This methodology allowed the team to build, test, and refine specific application modules—such as API communication, layout building, and UI rendering—step-by-step. The entire project lifecycle was categorized into five major development milestones.

4. MILESTONE MAPPING & EXECUTION

Milestone 1: Foundational Setup & Architecture

- **Primary Goal:** To finalize the technology stack and configure the working environment.
- **Key Activities:** Evaluated and selected Gemini 1.5 Flash (for structured JSON generation) and Gemini 1.5 Pro (for narrative text). Chose Stable Diffusion v1.5 for image rendering. Established a secure Python virtual environment and drafted the MVC (Model-View-Controller) directory structure.

Milestone 2: Backend AI Module Creation

- **Primary Goal:** To write the core Python scripts responsible for external AI communication and file formatting.

- **Key Activities:** Developed standalone modules like `gemini_handler.py` to fetch story outlines and dialogues. Created `image_handler.py` to request and download comic illustrations. Built the `pdf_compiler.py` utilizing the FPDF library to calculate layout dimensions for the final document.

Milestone 3: FastAPI Routing Integration

- **Primary Goal:** To construct the web server logic that bridges user inputs with AI modules.
- **Key Activities:** Authored `main_routes.py` to handle frontend POST requests. Sequenced the execution pipeline (Outline → Story → Images → Layout → PDF). Implemented comprehensive exception handling to manage API timeouts gracefully.

Milestone 4: UI/UX Frontend Design

- **Primary Goal:** To design the user-facing web pages ensuring an intuitive experience.
- **Key Activities:** Designed the primary `index.html` capturing user parameters (Character, Tone, Style). Developed the dynamic Jinja2 template (`preview.html`) configured to loop through backend data and render the comic panels sequentially on the browser.

Milestone 5: System Testing & Final Deployment

- **Primary Goal:** To perform end-to-end validation and ensure production readiness.
- **Key Activities:** Hosted the application locally using the Uvicorn server. Conducted rigorous testing on prompt accuracy, API failover speeds, and verified that the compiled PDF files generated without layout distortion.

5. ANTICIPATED CHALLENGES & CONTINGENCIES During the planning stage, the team identified potential technical bottlenecks and engineered proactive solutions:

- **Challenge 1: Cloud API Rate Limiting or HTTP Errors.**
 - *Contingency:* Built specific try-except blocks in the backend to detect 402/503 errors and instantly switch between backup image generation engines.
- **Challenge 2: Complete Network Blockage (e.g., DNS Resolution Failures).**
 - *Contingency:* Programmed a local PIL-based fallback script that instantly generates a yellow "Comic Caption" placeholder image if all internet APIs fail, ensuring the PDF compiler never crashes.

6. WORKFLOW ALLOCATION To maximize efficiency, technical responsibilities were strategically divided across the team. Specific focus areas included backend API integration, frontend Jinja2 templating, PDF layout arithmetic, and quality assurance network testing.